

Clarifying Questions & Problem Restatement

Northwind Systems — Fintech Onboarding & KYC

Workflow Characteristics

Onboarding runs for hours-to-days, depends on external KYC checks plus manual compliance review, and sometimes stalls. Policy changes can land while workflows are already in flight.

Assumptions If Unanswered

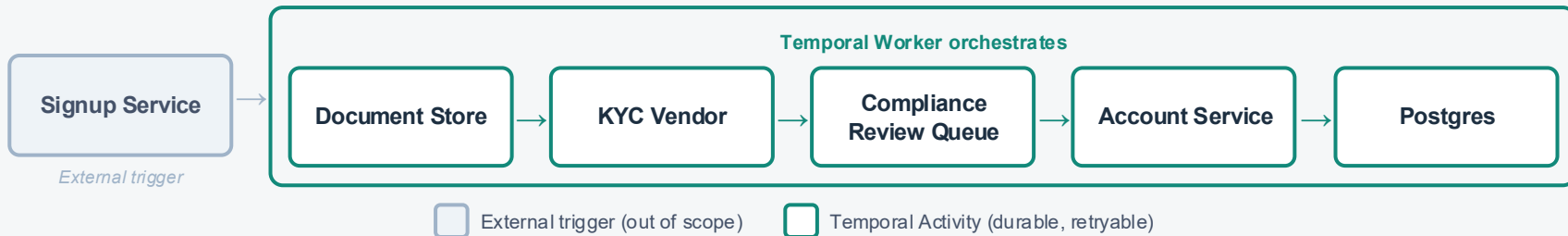
- KYC vendor is synchronous, not guaranteed idempotent
- 48-hour review SLA before escalation
- Moderate volume — hundreds of applications/day
- Net-new applications only for first release

Clarifying Questions Asked First

- What does "stuck" look like today — lost state, missing retries, or a visibility gap?
- What does the compliance review queue look like, and what happens if a reviewer is out for two days?
- If the KYC vendor is down for two hours, do we wait-and-retry or fail and notify the customer?
- Has a policy change ever broken an in-flight onboarding process before?
- Which team would own this workflow — signup, compliance, or a platform team?

Workflow Design — One Path, Clear Boundaries

Application Submitted → KYC Checks → Manual Review → Account Activated



KEY DESIGN DECISIONS

Timers

`Workflow.await(48h)` blocks without polling. Timer fires → escalates to a senior reviewer.

Human-in-the-Loop

`@SignalMethod reviewDecision()` — reviewer hits HTTP endpoint; Temporal Signal unblocks the wait.

Safe Evolution

`Workflow.getVersion()` branches old vs. new policy — in-flight executions keep their original logic.

Audit Trail

`onboarding_audit` written at every step. Temporal history = complete durable record for regulators.

Visibility

Temporal Web UI for ops; `SELECT * FROM onboarding_audit WHERE workflow_id = ?` for compliance.

Failure Handling, Idempotency & Ownership

Retries, timeouts, audit, and visibility

Concern

Temporal Mechanism

Retries + backoff

ActivityOptions retry policy — up to 5 attempts with exponential backoff for the KYC vendor call, 3 for everything else.

Long external waits

Activity heartbeat + StartToCloseTimeout / ScheduleToCloseTimeout — Temporal detects a hung vendor call without polling.

Idempotency

workflowId + activityId as the dedupe key; Postgres ON CONFLICT DO NOTHING makes every activity retry safe.

Audit trail

Every activity appends to onboarding_audit — a durable, regulator-visible history of the full lifecycle.

Operator visibility

Temporal Web UI for execution timeline and retries, plus a live SQL audit query for the business view.

Ownership

Platform team owns Temporal infra; compliance team owns review policy; signup team owns the activities.

Demo — One Success Path, One Failure Path

Live Workflow prototype: Java App + Temporal Java SDK + Postgres audit trail

Scenario A — Happy Path (with retries)

- KYC check fails on attempts 1–2 (simulated transient error), passes on attempt 3
- Workflow is resilient to infra failures
- Temporal Web UI shows the automatic retry — no custom code, no polling
- Compliance reviewer approves via curl → POST /review/{id}/approve OR /review/{id}/reject
- Account activated Or Application rejected
- Full audit trail written to Postgres

Scenario B — KYC Rejection (failure path)

- KYC_FORCE_FAIL=true — vendor returns false, no exception, no retry
- Workflow auto-rejected due to SLA breach (Timeout)
- Workflow reads false and calls rejectApplication() immediately
- Clean terminal REJECTED state with an accurate audit trail
- Shows a deliberate business failure path — not a crash, not a retry

Observability and Audit Trail

- Temporal Web UI shows full workflow life-cycle (execution history, retries, pending signals)
- Postgres DB: Shows full regulator friendly audit trail (SELECT * FROM onboarding_audit WHERE workflow_id = ?)

Rollout Plan — First 30–60 Days

Phased adoption

DAYS 1–30

Parallel Run and Observe

Temporal handles net-new applications only. Old system keeps running in-flight ones. Single shared namespace and task queue. Observe KYC and review queue response times.

DAYS 30–60

Harden, Expand and Decide Cutover

Add alerting on stalled workflows and confirm vendor idempotency keys. Increase volume, add a second escalation tier, and decide go/no-go on migrating remaining in-flight applications.

Risks

- KYC vendor may not provide idempotency keys today
- Reviewer queue has no SLA currently — escalation policy is new
- Volume spikes (e.g. campaigns) untested against worker sizing

Next Checks Before Production

- Load test with realistic and peak application volume
- Confirm namespace / task-queue sizing and worker autoscaling
- Security review of the ReviewServer signal endpoint (auth)